

XRD and Excel programming

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Before you start Programming

Plan out your programming

- Take more planning time upfront for what you need for Statements, DOPT Listings, Case Memo, and IPVFB
- Consider Statements
 - Before programming, ask what steps need to be shown on statements?
 - Separate what needs to be shown into different fields
- Consider DOPT Listings
 - Can a long formula be separated into multiple fields for ease of DOPT Listings programming and Peer Review?

Plan out your programming (continued)

- Consider IPVFB
 - Certain data needs to be populated in certain fields
 - See IPVFB Training from October of 2014, ADT results, and ACT User Group Minutes
- Consider Case Memo
 - Which fields do you need to add up?
 - Rounding errors should only be apparent in 4022(c)

Plan out your programming (continued)

- If you consider all programming standards at the beginning, you can program them in as you go
- If you try to program all standards in at the end, it may result in lost time trying to work additional programming into the current formulas

Plan out your groups (continued)

- Unique Groups of Participants
 - Usually easier to start programming a group that contains most participants in the same type of situation
 - Small groups with unique provisions then can be separated into a different group
 - Limits the amount of IF statements in each cell making it easier to identify/correct issues and to review
 - Examples: Transfers, QDROs, etc.

Use pre-defined fields

- Don't default to a user defined field
- ACT has many pre-defined fields which should be used, if applicable
- [Data Dictionary](#)

Questions?

XRD Only Programming Basics

What is XRD Only programming?

- Allows for calculating plan and termination benefits at different dates while programming the calculation once
- Used primarily for participants not in-pay status as of DOPT

Advantages of XRD Only programming

- Simplifies programming and reduces reference errors by using less columns in Excel
- Reduces chance for mistakes from having to program the same benefit calculation multiple times
- Simplifies review of programming
- Isolate dates for statement programming

Advantages of XRD Only programming

- Simplifies revisions as the correction only needs to be applied to one calculation
- Can reduce update time on large cases by focusing updates on individual dates
- Help pinpoint which calculation date is causing an error

XRD Only programming Scenarios

- XRD – Used for valuation liabilities and recalculation statements
- QPSA – Generally used to calculate XRD_MB_QPSA_TERM and LS_QPSA
- NRD/EURD/ERD – Used for statements for not in-pay participants
- RASD – Used to calculate a benefit for a participant eligible for a retirement date (excluding RBD) prior to BD Issuance date (Stale Date)
- DORNSF – Used to Calculate single form PACS optional forms for post-DOPT retirees
- DOR – Used to calculate married PACS optional forms and reconcile post-DOPT retiree benefits

XRD Only programming Scenarios (continued)

- You can program other dates or situations that you deem necessary (such as RBD).
- You can have up to 10 scenarios
- XRD Only Field Mapping Word Document

V1 Configuration for XRD Only Programming

- Each row must include C1.RETSTAT, C1.ID, C1.CUSNUM, and C1.CASE
- Name of each row before “I/O/B” is what will appear in dropdown box for retrieve group and update
- QPSA, DORNSF, and DOR have extra functionality with ACT programs
 - QPSA – If a recalculation with QPSA_INDICATOR = “Y” for an ID = “1” participant is run, the QPSA row will be called first by RunTime to recalculate LS_QPSA and then proceed with calculating the benefit using the XRD row
 - DORNSF and DOR – Automatically pull participants with DOR populated into the **BUCK** group

V1 Configuration for XRD Only Programming (continued)

- CALCULATION and CALC_INDICATOR fields are required for tabs with XRD Only programming
 - CALCULATION will automatically populate with the row name upon retrieving group or update
 - CALC_INDICATOR is populated depending on which ACT program produced the calculation:
 - “V” for Valuation
 - “R” for Recalculation.
 - Very useful to distinguish between valuation and statement programming (i.e. non-spouse beneficiaries and BFCFs)



V1 Configuration for XRD Only Programming (continued)

- V1 Early Setup Exercise

V1 Configuration for XRD Only Programming (continued)

I/O/B rows

- “I” indicates an input field. Data element is pulled from database upon calculation.
- “O” indicates an output field. Requires a formula and is updated back to the database upon an update in ValTool. Data element is recalculated in Runtime.
- “B” indicates a field that is an output field in ValTool and an input field in Runtime.
- “N” indicates a field that is calculated but not updated back to the database in ValTool. Useful for intermediate steps.
- “P” indicates a field that is an output on the first pass in Runtime and an input on the second pass when a married or non-spouse recalculation is run.



V1 Configuration for XRD Only Programming (continued)

XRD Only Programming Example

Questions?

Setting up your V1

Setting up your V1 sheets

- Setup clearly defined sheets that are well organized
- Common Sheets
 - Valuation – Programming sheet
 - Tables – includes factor tables, dates, and other references for programming
 - Summary – used for summarizing liabilities and calculating other values for case memo tables

Valuation Sheets

- Common valuations groups on most cases are In-Pay and Not In-Pay
- Separate unique groups into their own sheets for ease of programming and review
- Possible grouping scenarios for more complex plans
 - Group by plan provisions
 - Group by sub-plan

Tables and Dates Sheet

- Define and label all dates and tables clearly
 - Factor tables should have the basis in which they were generated labeled including PBGC mortality table name, if applicable
- Name ranges should describe the date or table
- J&S BFCF tables must have spouse ages from 0-100
- Remove unused tables and name ranges

Summary Sheets

- Include Case Memo Tables
 - Total Liabilities with Load (First page of Case Memo)
 - 4022(c) calculation (First page of Case Memo)
 - Asset Allocation (Section 17)
 - Benefit Statement and Retirement Statement Counts (Section 25)
 - Expense Load (Section 27)
 - Liability-Weighted Average Ages (Section 28)
 - Total Liabilities without Load (Section 28)
- Calculate 4044 and 4022(c) allocation ratios

Summary Sheets

1	GROUP w/ Load	Number	PV of Title IV	PV of UNGB	PVPBL	PV of 4022(c)	PVMB_TERM
2	Retirees	20	\$6,722,415	\$137,574	\$6,859,989	\$14,051	\$6,736,466
3	Beneficiary	0	\$0	\$0	\$0	\$0	\$0
4	Retirees & Beneficiaries subtotal	20	\$6,722,415	\$137,574	\$6,859,989	\$14,051	\$6,736,466
5	Separated Vesteds	72	\$2,335,956	\$12,064	\$2,348,020	\$0	\$2,335,956
6	Deferred Bene	0	\$0	\$0	\$0	\$0	\$0
7	Separated Vesteds subtotal	72	\$2,335,956	\$12,064	\$2,348,020	\$0	\$2,335,956
8	Active Vested	152	\$20,317,941	\$0	\$20,317,941	\$0	\$20,317,941
9	Non Vesteds	0	\$0	\$0	\$0	\$0	\$0
10	TOTAL	244	\$29,376,312	\$149,638	\$29,525,950	\$14,051	\$29,390,363

12	GROUP W/O Load	Number	PV of Title IV	PV of UNGB	PVPBL	PV of 4022(c)	PVMB_TERM	LS_TERM	TERM_LW_ANB	TERM_LW_XRA
13	Retirees	20	\$6,675,685	\$136,618	\$6,812,303	\$13,954	\$6,689,639	\$6,475,170	\$438,616,139	
14	Beneficiary	0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	
15	Retirees & Beneficiaries subtotal	20	\$6,675,685	\$136,618	\$6,812,303	\$13,954	\$6,689,639	\$6,475,170	\$438,616,139	
16	Separated Vesteds	72	\$2,319,719	\$11,980	\$2,331,699	\$0	\$2,319,719	\$2,285,362	\$131,203,327	\$139,678,831
17	Deferred Bene	0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
18	Separated Vesteds subtotal	72	\$2,319,719	\$11,980	\$2,331,699	\$0	\$2,319,719	\$2,285,362	\$131,203,327	\$139,678,831
19	Active Vested	152	\$20,176,709	\$0	\$20,176,709	\$0	\$20,176,709	\$20,199,128	\$1,197,902,162	\$1,240,356,983
20	Non Vesteds	0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
21	TOTAL	244	\$29,172,113	\$148,598	\$29,320,711	\$13,954	\$29,186,067	\$28,959,660	\$1,767,721,628	\$1,380,035,814

23	The Expense Load Was Calculated as Follows:		RET=1	RET=2 or 3	RET=4	TOTAL
24	(1) Number of Annuity Participants:		20	224	0	244
25	(2) PVPBL without Load (annuity participants only):		\$6,812,303	\$22,508,408	\$0	\$29,320,711
26	(3) Initial Annuity Interest Rate:					2.50%
27	(4) $p = 1\% + [(3) - 7.50\%] / 10 =$					0.00500
28	(5) Amount of Load: $\$10,000 + [(2) - \$200,000] \times (4) + \$200 \times (1) =$					\$204,404
29	(6) Loading Percentage: $(5) / (2) =$					0.0070

31	A) Present Value of Title IV Benefits:	\$29,376,312
32	B) Final Assets Available for Allocation:	\$24,170,365
33	C) Unfunded Guaranteed Benefits:	\$5,205,947
34	D) Unfunded Non-Guaranteed Benefits:	\$149,638
35	E) Unfunded Benefit Liabilities:	\$5,355,585
36	F) SPARR:	9.39%
37	G) 4022(c) Amount:	\$14,051
38	H) PV of Termination Benefits:	\$29,390,363

	TERM_LW_ANB	TERM_LW_XRA
Retirees/Beneficiaries	65.6	N/A
Separated Vesteds	56.6	60.2
Active Vesteds	59.4	61.5
Non-Vesteds	0.0	0.0

\$0

PC3	UNGB PC3	PC4	PC5	PC6	Total UNGB
\$6,696,229	\$0	\$26,186	\$137,574	\$0	\$137,574
\$0	\$0	\$0	\$0	\$0	\$0
\$560,259	\$0	\$1,775,697	\$0	\$12,064	\$12,064
\$0	\$0	\$0	\$0	\$0	\$0
\$12,310,043	\$0	\$8,007,898	\$0	\$0	\$0
\$0	\$0	\$0	\$0	\$0	\$0
\$19,566,531	\$0	\$9,809,781	\$137,574	\$12,064	\$149,638

LOAD		TRUE
PC3 4044 Percentage	100.0000%	TRUE
PC3 4022(c) Percentage	100.0000%	TRUE
PC5 4044 Percentage	0.0000%	TRUE
PC5 4022(c) Percentage	10.2134%	TRUE
PC6 4044 Percentage	0.0000%	TRUE
PC6 4022(c) Percentage	0.0000%	TRUE

Summary Sheets

- Sum BS_IND, BR_IND, and MAN_STMT_IND

Group	Number of BSRS Statements	Number of Manual Statements	Number without Statements	Total
Retirees and Beneficiaries	201	1	2880	0
Separated Vesteds	9625	0	98	0
Active Vesteds	155	0	0	0
Non-Vested Participants	12	0	0	0
Alternate Payees not in the Participant Count Above	0	0	0	0
Total	9993	1	2978	12972
Group	Number of BSRS Recalculations	Number of Manual Recalculations	Number without Recalculations	Total
Retirees and Beneficiaries	66	0	3016	3082
Separated Vesteds	8536	0	1187	9723
Active Vesteds	135	0	20	155
Non-Vested Participants	0	0	12	12
Alternate Payees not in the Participant Count Above	0	0	0	0
Total	8737	0	4235	12972



How to sum

- Use DSUM or COUNTIFS functions
- (Excel Example)



Questions?

XRD Programming

Best Practices

- Program Left to Right
 - Only reference fields preceding calculated field
 - Organizing the v1 helps with ease of programming DOPT Listings and Peer Review

Best Practices (Continued)

- General Rounding

—All fields appearing on statements or DOPT Listings should be rounded

Field Type	Decimal Rounding
Monthly Benefits	2 decimals
Plan Factors (ERF, LRF, BFCF, conversion, etc.)	Plan practice, else 4 decimals
Present Values	0 decimals
Load Percentage	4 decimals
Ratios (4044, 4022(c), Guaranteed, etc.)	6 decimals

Best Practices (Continued)

- Liabilities Rounding
 - Try to multiply as little as possible
 - Multiplying when you can add already calculated fields resulting in more \$1 rounding error
 - Multiply only in the PC fields and 4022(C)
 - Add existing PC calculations for Title IV, UNGB, PVPBL, and TERM liabilities

Best Practices (Continued)

- Survivor and QPSA Monthly Benefits Rounding
 - For ID=2 or QPSA beneficiaries
 - Do not include the multiplication of SPC in the calculation of the participant's benefit.
 - Statements need the rounding done in separate fields
 - Participant's MB = $PA_AMB \times ERF_XRD \times LRF_XRD \times _BFCF_XRD$
 - Beneficiary's MB = Participant's MB x SPC

Best Practices (Continued)

- Use the most recent versions of ACT functions
 - i.e. NPVF2, XRD2, XRA2...

Data Inputs

- Participant, Spouse, Non-Spouse, Alternate Payee Info
- Employment History field
 - Status dates (DOH, DOP, DOTE, DOBS1, DORH1, etc.)
 - Location/division (EMPLOYEE_CODE1, EMPLOYEE_CODE1_DESC, etc.)
- Benefit Calculation fields, if applicable
 - Salaries, Benefit Rates, Services, AMBs, Etc.
- QDRO Information, if applicable
- Indicators
 - Any standard or user-defined fields to assist with programming

AMB Calculation

- If calculating or recalculating any AMBs, include fields included in calculating the benefit
 - CS, VS, ACOMP, BR, AMB, etc.
- If AMB calculation is too cumbersome, consider calculating them on a separate sheet or file and loading them as inputs on the valuation sheet.

Key Dates and Ages

- NRD
- Early Unreduced Retirement Ages and Dates
- Early Retirement Ages and Dates
- RBD
- BDERDLEV and BDEUAS
- RETRO_DOR

Key Dates and Ages: NRD (Continued)

- RETDATE function
 - RETDAT(DOB, Retirement Age, FOMA, Anniv)
 - The RETDATE function will accept and use a non-integer value for the retirement age parameter. For example if the user enters 70.5 for the retirement age, the function will return the first of the month on or after the 70.5 birthday. Note the value entered in the retirement age parameter is rounded to the nearest completed years and months.
 - Not meant to replace the NRD function, but useful in calculating NRDs at non-integer ages

Key Dates and Ages: NRD (Continued)

- RETDATE function

—FOMA parameter:

- 0 is the first of the month on or following (default value)
- 1 is the first of the month following
- 2 is the first of the month on or before
- 3 is the plan anniversary nearest
- 4 is the first of the month nearest

—Anniv parameter:

- Must be populated when FOMA = 3. Format should be in MMDD, e.g., “0301” for a March 1st plan anniversary

Key Dates and Ages: Early Unreduced Retirement (Continued)

- PL_EURD – Plan's Earliest Unreduced Retirement Date
 - Use NRD or RETDATE functions
- PL_EURA – Plan's Earliest Unreduced Retirement Age
 - Age nearest birthday on PL_EURD
- EURD – Plan's Earliest Unreduced Retirement Date capped at DOPT
 - MAX(PL_EURD, FOMF DOPT)
- EURA_XRA – Plan's Earliest Unreduced Retirement Date as of DOPT
 - Should equal ANB on DOPT if PL_EURD is before DOPT

Key Dates and Ages: Early Retirement (Continued)

- PL_ERD and PL_ERA
 - Make sure you incorporate PBGC EPRD
- ERD – Plan's Earliest Retirement Date capped at DOPT
 - MAX(PL_ERD, FOMF DOPT)
- ERA_XRA – Plan's Earliest Retirement Date as of DOPT
 - Should equal ANB on DOPT if PL_ERD is before DOPT

Key Dates and Ages: RBD (continued)

- The field RBD_DT must be populated in the database with the **participant's** required beginning date for every record where BR_IND is set to 'Y' (i.e., enabled for recalculation).
- Beneficiary's RBD must be stored in a user-defined field. Runtime automatically calculates beneficiary's RBD for a QPSA recalculation statement.

Key Dates and Ages: RBD (continued)

- RBD function
 - RBD(RETSTAT, DOPT, Stype, DOB, DOTE, DOD, SOwner, NonSpouseInd)
 - The function has been updated for the new Age 72 RBD policy
 - Stype* is the calculation type. Valid values are "P" for participant's required beginning date and "B" for beneficiary's required beginning date.
 - NonSpouseInd* is the indicator for a non-spouse beneficiary (NON_SPOUSE_BENF field). Valid values are "Y" and "N". This is an optional parameter. Note, when the *Stype* parameter is set to "B" and this parameter is set to "Y", the function will return the earlier of the participant's required beginning date and the 12/01 of the calendar year following the calendar in year in which the participant dies.

Key Dates and Ages: BDERDLEV and BDEUAS (Continued)

- Used for statements for participant not yet in-pay status
- Normally no earlier than the projected BD issuance date; often referred to as the Stale Date or Current Date.
- Should only be set to a date prior to Current Date if RBD is prior to the Current Date. Use RETRO_DOR for other retractive retirement scenarios.

Key Dates and Ages: Retroactive Retirement (Continued)

- RETRO_DOR – The earliest retroactive retirement date to which a payee is entitled when the BD is issued.
- Used for retroactive scenarios including EURD, QPSA, Title IV
- Not used for the retroactive RBD scenario

XRA Assumptions

- ATM Method 1 (4044.47) – Special Rule for Facility Closing
 - The facility at which a participant is or was employed is in the process of being permanently closed on DOPT or was permanently closed within one year before DOPT.
 - The participant is either employed at the facility on DOPT or left employment at the facility less than one year before DOPT.
 - BL = HIGH and XRA = ERA

XRA Assumptions

- ATM Method 2 (4044.46) – Facility Closing does not apply and participant does not need to leave employment
 - BL = HIGH and XRA = ERA

XRA Assumptions

- ATM Method 3 (4044.45) – Facility Closing does not apply and participant does need to leave employment
 - BL = LOW, MED, or HIGH

XRA Assumptions

- Separated Vested for Method 2 and 3
 - BL = HIGH
 - Rare exception for which the plan provides further restrictions on employers other than the plan sponsor. i.e. working in same industry

XRA Assumptions

- BL function
 - BL(DOPT, DOB, DOTE, ishut, iquit, NOIT, RETSTAT, EURA, GMBNRA_VB_NSF)
 - ishut is indicator of facility shutdown (“Y” or “N”)
 - Note: Be careful with this indicator in plans involving asset sales. The company may be shutdown, but the facility may still be open.
 - Iquit is indicator of whether an individual participant has to quit to retire (“Y” or “N”)
 - In plans where the facility is not closed, a participant that has terminated prior to DOPT needs this indicator set to “N” because they have already terminated or the function will treat them as they have not terminated.



XRA Assumptions (continued)

- BL function (continued)
 - The BL function does not automatically return “IMM” for ID = 2 records or participants past age 65 on DOPT.
 - IF (OR(ID = “2”, ANB>=65, “IMM”), BL(DOPT, DOB, DOTE, ishut, iquit, NOIT, RETSTAT, EURA, GMBNRA_VB_NSF))

XRA Assumptions (continued)

- An incorrect setting in the IQUIT parameter can lead to the BL function returning incorrect results. If the XRA is being determined under the shutdown rules, the ISHUT parameter should be set to 'Y' and the **IQUIT parameter should be set to 'N'**, i.e., you should never have IQUIT set to 'Y' when ISHUT is set to 'Y'. This is because if the Special Facility Closing Reg 4044.57 applies, XRA is never dependent on whether a participant needs to terminate DOPT employment to receive a retirement benefit.

XRA Assumptions (continued)

- If the XRA is being determined under the on-going rules, it is unusual to have the IQUIT parameter set to 'Y' for a separated vested record. Setting IQUIT to "Y" for a separated vested participant means that he or she is working for a related employer as of DOPT and, under the terms of the plan, must quit from that employment in order to receive a retirement benefit from the plan trustee by the PBGC.

XRA Assumptions (continued)

- XRA2 function
 - XRA2(BL, ERA, NRA, DOPT, DOB, SDOB, RETSTAT, ID, FOMA, Anniv, DOP)
 - NRA parameter must be populated with the earliest unreduced retirement age as of DOPT
 - If the participant is eligible for an unreduced retirement at an age that is less than the normal retirement age, enter the earliest unreduced retirement age as of DOPT in the NRA parameter.
 - If the participant is past the normal retirement date at DOPT, enter the age at DOPT in the NRA parameter even if this is greater than the normal retirement age.
 - FOMA, Anniv, and DOP parameters are all optional

XRA Assumptions (continued)

- **Note** ID = '4'(Alternative Payee) is not supported for the *ID* parameter because the expected retirement date for an alternate payee record that is deferred at DOPT is equal to the expected retirement date for the participant (assuming the participant was also deferred at DOPT). Therefore, the user should compute the expected retirement age and expected retirement date for the participant using the BL, XRA2, and XRD2 functions and then use the SXRA2 function to compute the expected retirement age for the alternate payee. The expected retirement age for the alternate payee should be stored in the XRA field in the alternate payee record (ID = '4').

XRA Assumptions (continued)

- XRD2 function
 - XRD2(RETSTAT, DOPT, BDERD, XRA, DOB, FOMA, Anniv, DOP)
 - FOMA, Anniv, and DOP parameters are all optional

XRA Assumptions (continued)

- SXRA2 function
 - SXRA2(RETSTAT, DOPT, VAL_SDOB, XRD, PL_ERD)
 - When running calculations in the BSRS/BET program, non-spouse beneficiaries can be entered that are born after DOPT and therefore VAL_SDOB will be after DOPT resulting in SXRA2 returning an error.
 - To avoid the error, set the I/O/B setting for the SXRA field to B so that the formula for the field is not evaluated on the ARD calculation passes. Note, this solution is predicated on an assumption that SXRA is not used in the calculation of the ARD benefit.

Plan Benefit Calculation

- Order of plan benefit calculation
 - Eligibility for participation
 - Service calculation
 - Specific number of hours in a defined period
 - 1,000-hour rule
 - Assume 2,080 hours per PY if hours data unavailable
 - Common function: SERVYCA, SERVYPA
 - Elapsed Time
 - Typically measure from DOH or DOP to DOTE
 - Fractional years usually rounded to nearest month, but it depends on plan practice
 - Common function: COMPM, COMPYM

Plan Benefit Calculation

- Order of plan benefit calculation
 - Accrued Monthly Benefit
 - Vested Benefit
 - Early Retirement Factors
 - Late Retirement Factors
 - Benefit Form Conversion Factors

Application of PBGC Limits

- BPD

- Types of BPD cutbacks

- Service
 - Compensation
 - Eligibility
 - Unreduced benefits, different factor tables, disability, etc.

Application of PBGC Limits

- BPD (continued)
 - Methods to calculate BPD service
 - Use actual hours or elapsed time prorating hours in year of BPD.
 - Calculate estimated hours or elapsed time between BPD and DOTE
 - Methods to calculate BPD compensation
 - Use actual salaries prorating salary in year of BPD
 - Calculate estimated compensation given known data



Application of PBGC Limits

- BPD (continued)
 - Methods to calculate BPD eligibility
 - Calculate age requirements as of BPD
 - Use BPD service
 - Methods to apply BPD cutback
 - Calculate BPD benefit using plan formula, BPD calculated values, and BPD factors
 - If you cannot match a plan benefit exactly, sometimes a ratio of BPD/DOPE values is appropriate



Application of PBGC Limits

- Accrued-at-Normal (AAN)
 - PBGC does not guarantee benefits in excess of AAN; i.e., PC5
 - Typically applies to plans with Supplemental Benefits; e.g., Steel and Automotive Plans
 - The AAN is the benefit calculated at NRD in the SLA form
 - Not applicable at all retirement dates; i.e., may only apply at a future date and not XRD
 - Disabled Participants with supplements are not subject to AAN limitation

Application of PBGC Limits

- Accrued-at-Normal (AAN) (continued)
 - Supplement amounts should be calculated in the pre-defined SUPP fields
 - XRD_SUPP1_TITLE_IV, XRD_SUPP1_4022C, XRD_SUPP1_TERM, etc.
 - Make sure to Program associated supplement start and end date fields
 - BD_XRD_SUPP1, ED_XRD_SUPP1, BD_ERD_SUPP1, ED_ERD_SUPP1, etc.
 - Make sure to program associated supplement form fields
 - i.e. FORM_CODE_SUPP1, FORM_CODE_SUPP2, FORM_CODE_SUPP3
 - “0” means SLA
 - “1” means same form as primary level life benefit

Application of PBGC Limits

- Maximum Guaranteed Benefit

—MAXLIMPPA function

- MAXLIMPPA(*DOPT, BPD, XRD, DOB, DOD, SDOB, SDOD, BDOB, BDOD, Nonspind, FORM, SPCT, MTHS, Hcavg, retstat, ID, Disabmil, CertainBen, LevelizedBen, SurvivorBen*)
- Should be used instead of the MAXLIM function
- The user should populate FORM, SPC, and MTHS based on the participant/AP's original form of annuity
- For a participant who commenced his/her benefit before dying, this is the form of annuity for the participant's benefit even if the participant died before DOPT (i.e., ID = '2')
- For a participant who died before retiring and before DOPT (i.e., ID = '2' and beneficiary is receiving a QPSA benefit) this is the joint-life form of annuity upon which the QPSA is based, which is generally the same as the QJSA



Application of PBGC Limits

- Phase-In

- PHASEINPCT Function

- PHASEINPCT(EffectiveDate, AdoptedDate, DOPT, BPD, UCEBdate, SEPPAA)
 - Returns the phase-in percentage as a decimal value based on the inputs specified
 - Need to round the field to 2 decimals
 - ECEBdate parameter is an optional parameter for the date of the UCEB (e.g., plant shutdown down date for a shutdown benefit) and should be populated only when computing the phase-in percent of a UCEB.
 - SEPPAA parameter can be set to 'Y' if the phase-in is to be performed using the pre-SEPPAA rules



Application of PBGC Limits

- Phase-In

- PHASEIN Function

- PHASEIN(percent, benefit)
 - Returns the phased-in benefit amount for the benefit at the phase-in percent
 - Need to round the field to 2 decimals

Present Value Factors for liabilities

- See July 2020 Factors Training
- Use DOPT to look up interest rates and mortality within the NPVF2 function
 - For checking purposes you can enter a range of interest rates and the mortality tables
- When using DOPT as an input to look up interest rates and mortality, PHS must be used as a parameter in the NPVF2 function

Present Value Factors for liabilities (continued)

- Disability Mortality

- Disability mortality should be based on the plan's definition and not plan practice
 - i.e. if the plan did not require SSA disability but in plan practice they used SSA disability as evidence, mortality should be based on SSA disability not being required, PHS = "1"
- Disabled participants age 65 and older as of DOPT should be valued as healthy
 - PHS should be changed to "0" or an IF statement included in the PHS parameter
 - NPVF2("MP", "JSC", "N",DOPT,PSEX,ANB,XRA,SXRA,SPC,MTHS,TEMP,DOPT,DOPT, "P",IF(ANB>=65, "0",PHS))

Present Value Factors for liabilities (continued)

- Valuing temporary supplements
 - Since the supplement is a temporary benefit, it needs to be valued with a separate PVF
 - PVF_SUPP1-PVF_SUPP3 are defined fields for supplement present value factors
 - NPVF2 function:
 - All inputs same as PVF_LEV_ANN except for “TempPeriod”
 - “TempPeriod” input equals months from supplement start date to supplement end date

Application of Multiple PBGC Limits: Valuing Supplements

- Present Value of Benefit with supplement =

Lifetime Monthly Benefit x PVF_LEV_ANN x 12 + Supplement x PVF_SUPP1 x 12

$$\text{PC4_ANNUITY_PVMB} = \text{PC4_LEVEL_MB} \times \text{PVF_LEV_ANN} \times 12 + \\ \text{PC4_SUPP1_MB} \times \text{PVF_SUPP1} \times 12$$

$$\text{PC5_ANNUITY_PVMB} = \text{PC5_LEVEL_MB} \times \text{PVF_LEV_ANN} \times 12 + \\ \text{PC5_SUPP1_MB} \times \text{PVF_SUPP1} \times 12$$

Priority Categories

- PC3

- Plan benefits that are in-pay, or could have been in-pay three years prior to DOPT based on Plan provisions five years prior to DOPT.

- PPA 2006 revised the PC3 rules for plans that terminated in bankruptcy on or after 09/16/2006 to use BPD instead of DOPT

- PBGC uses a two-step process to assign benefits to PC3:

- 1. Identify participants actually receiving or eligible to receive Plan benefits at DOPT/BPD - 3
 2. Calculate the PC3 benefit

Priority Categories

- PC3

- Identify participants

- Participants, Beneficiaries, or Alternate Payees receiving benefits at DOPT/BPD - 3 are PC3 eligible
 - Participants that reached their EPRD on or before DOPT/BPD - 3 are PC3 eligible
 - Beneficiary or Alternate Payee in pay at DOPT where the participant was not in pay at DOPT/BPD – 3 are PC3 eligible if the participant reached their EPRD on or before DOPT/BPD - 3

- PC3 eligibility is based on the exact day the participant reached their EPRD, not the first of the month following. See the following example:

Priority Categories

- PC3

—PC3 eligibility Example:

DOPT: 01/10/2018

DOPT/BPD - 3: 01/10/2015

Participant's EPRD: 01/05/2015

This participant is eligible for PC3 because he reached EPRD prior to DOPT/BPD - 3 even though he couldn't have received a benefit prior to DOPT/BPD - 3

Priority Categories

- PC3

—Programming PC3 eligibility:

- Use defined field PC3_ELIG_IND
- In general, set PC3_ELIG_IND = “Y” if:

DOR $\leq$ DOPT/BPD – 3 or
Participant’s EPRD $\leq$ DOPT/BPD – 3

Priority Categories

- PC3

- Calculate PC3 Benefits

- PC3 Benefits are calculated at the PC3 Calculation Date:
 - PC3 Calculation Date for participants, beneficiaries, and alternate payees in pay at DOPT/BPD – 3 is the Annuity Starting Date
 - PC3 Calculation Date for anyone not in pay at DOPT/BPD – 3 is the first of the month coincident or following DOPT/BPD – 3
 - PC3_LEVEL_MB is the monthly benefit calculated at the PC3 Calculation Date, net of the monthly benefits in PC2 and PC1.

Priority Categories

- PC3

- Calculate PC3 Benefits

- PC3 Benefits are calculated at the PC3 Calculation Date:
 - PC3 Calculation Date for participants, beneficiaries, and alternate payees in pay at DOPT/BPD – 3 is the Annuity Starting Date
 - PC3 Calculation Date for anyone not in pay at DOPT/BPD – 3 is the first of the month coincident or following DOPT/BPD – 3
 - PC3_LEVEL_MB is the monthly benefit calculated at the PC3 Calculation Date, net of the monthly benefits in PC2 and PC1.



Priority Categories

- PC3 Exercise
 - Calculate the ages at PC3 date for eligibility and factors for the two sample participants. See “PC3 Exercise” in the “Sample V1 and Exercises.xlsx” spreadsheet
 - Early retirement eligibility is age 55 after attainment of age 55 with 10 years of vesting service.
 - Age is defined as age nearest birth

Priority Categories

- PC5
 - Nonforfeitable plan benefits with respect to the participant
 - PC5 generally results from PBGC limitations (BPD, AAN, MIL, PI)
 - Assets are allocated to PC5 starting with the total value of benefits on DOPT-5 not covered by earlier priority categories
 - Allocation of remaining assets (if any) progresses in the order the amendments
 - PC5_LEVEL_MB is used if only one source of liability
 - If more than one source (different amendments) the amounts are split into PC5A_LEVEL_MB through PC5E_LEVEL_MB

Priority Categories

- PC6
 - All other “accrued” benefits
 - PC6 commonly contains liability from non-vested benefits and death benefits in excess of the QPSA
 - PC6_LEVEL_MB is the monthly benefit calculated at XRD, net of the monthly benefits in all other priority categories.

Priority Categories

- Calculate PC fields through Total
 1. PC#_ANNUITY_PVMB
 2. PC#_QPSA_PVMB
 3. PC#_LOAD_PVMB ($[1 + 2] \times \text{LOAD_PCT}$)
 4. PC#_Total_PVMB ($1 + 2 + 3$)

Note: There are no QPSA fields for PC1 and PC2. Instead there are LSDB fields.
i.e. PC2_LSDB_PVMB

Final Liabilities

- Liabilities Order
 - Block the five memo liabilities into like fields (Title IV, UNGB, PVPBL, 4022(c), TERM)
 1. No QPSA, No Load
 2. QPSA
 3. No Load (1 + 2)
 4. Load
 5. Total Liabilities (3 +4)

Final Monthly Benefits

- Can be placed either before or after Final liabilities depending on if any benefit is derived from the present values

Lump Sums

- Use DOPT to look up interest rates and mortality within the NPVF2 function
- LS_TERM
 - Should be the lump sum for the record being valued. i.e. participant, beneficiary, or alternate payee
- LS_QPSA
 - Calculated only for participants due a QPSA benefit

Survivor and QPSA monthly benefit fields

- Survivor Fields should be based on the plan's QJSA
- QPSA Fields should be based on the plan's QPSA
- Best practice is to calculate Survivor and QPSA fields for all participants
- Participant's with post-DOPT deaths will have errors in ADT if QPSA and Survivor fields are not populated

PACS

- The fields PBGC_OPT_SLA, PBGC_OPT_5CC, etc. is where PBGC optional forms of annuity are calculated. They are referenced on the OFA generated by BSRS in the standard p1 configuration file
- PBGCOFA function
 - PBGCOFA(DOPT, DOB, SDOB, BDOB, NonSpind, DOR, AnnTp, BenForm, SrvPct, MthsCert, MthsTemp, MthlyBen, PopupBen, PBGC SLA, PBGC Form, SrvBenInt1, StrtDtInt2, SrvBenInt2, StrtDtInt3, SrvBenInt3, FreeCCMthlyBen, FreeCCEndDate, EECBal, PMedSuppDt, PMedSuppAmt, BMedSuppDt, BMedSuppAmt, BMedSuppInd, AAN, TmpSupp, SuppEnd, CertDeath)

PACS (Continued)

- PBGC_OPT_SLA and PBGC_OPT_##CC:
 - The DOB input should be set to APDOB for ID=4, SDOB for QPSA_INDICATOR=Y, else set it to the participant DOB
 - SDOB and BDOB are unused and can be left blank. NonSpind should be “N”.
 - The DOR input should be set to the actual date of retirement. For BSRS, this populates in XRD. For reconciliation, it is in DOR.
 - AnnTp is always set to “MP”
 - BenForm generally always set to “SLA”. SrvPct is generally set to 0. MthsCrt is set to MTHSNSF for participants and APs, and MTHS_QPSA for QPSAs. MthsTemp is generally 0.
 - MthlyBen is the benefit in the normal single/QPSA form. PopBen and PBGCSLA are unused
 - PBGCForm is the PBGC optional form being computed. “SLA”, “5CC”, “10CC”, “15CC”

PACS (Continued)

- PBGC_OPT_JS## and PBGC_OPT_POPUP:
 - The DOB input should be set to participant DOB. SDOB input should be set to SDOB. BDOB input should be set to BDOB. NonSpind should reference NON_SPOUSE_BENF.
 - The DOR input should be set to the actual date of retirement. For BSRS, this populates in XRD. For reconciliation, it is in DOR. AnnTp is always set to “MP”
 - BenForm generally always set to “JSC”. SrvPct is set to SPCNMF. MthsCrt is set to MTHSNMF. MthsTemp is generally 0.
 - Populate MthlyBen the normal married form amount. Populate PopBen and PBGC_SLA with the amount calculated in PBGC_OPT_SLA
 - PBGCForm is the PBGC optional form being computed. “JS50”, “JS75”, “JS100”, “JSPOP”
 - For BSRS, UD fields should be used to calculate the survivor portion of these.

JSPOPUP

- When the form of annuity is a J&S Popup, the JSPOPUP field is populated with the amount of the increase to the primary annuitant's benefit when the contingent annuitant dies before the primary annuitant.
- The field JSPOPUP is only required for expected retirement benefits and is not required for benefits in pay.
- For records deferred at DOPT, JSPOPUP should always be populated with the amount based on the expected retirement benefit (leave unpopulated if the expected retirement form of annuity is not a J&S Popup) even if the actual retirement benefit has commenced.

JSPOPUP (continued)

- For records in pay at DOPT, JSPOPUP does not have to be populated (not a problem if it is populated).
- If the participant is entitled to multiple benefits that are administered separately (i.e., the multiple XRD benefit fields in the C9 table are populated), the JSPOPUP field only has to be populated if the benefit (both those in pay and those still deferred) with the largest DOPT value is still deferred and is a J&S Popup annuity.



Questions?

Programming at other dates

General tips for other Rows

- If different values are needed at different dates, be sure to use different fields so you don't overwrite values that are needed.
 - i.e. plan benefits, plan factors, UNGBs, etc.
- Fields that need to be recalculated but don't need to appear on statements can generally be set to “N” in the I/O/B row instead of using a user defined field and to avoid overwriting the XRD value
 - i.e. VAL_SDOB, PC MB allocation , etc.
- Fields that aren't recalculated can be changed to inputs (“I”) instead of outputs (“O”)
 - i.e. AMB, allocation ratios, key dates , etc.

QPSA Row

- The QPSA row is an optional row that can be included in the v1. This row will have additional functionality:
 - Runtime will calculate the QPSA row after the ARD Data Entry screen and prior to calculating the XRD row for a Retirement Statement
 - Field 'CALCULATION' will be set to 'QPSA'
 - Field 'CALC_INDICATOR' will be set to 'V' (provided this is correctly programmed in the v1)
 - The QPSA row will NOT be called for ID='2' Retirement Statements as well as Post-DOPT QPSA retirees

QPSA Row

- There are two primary reasons to include the QPSA row:
 - When a plan has a QPSA that is different from the QJSA
 - IPVFB requires all participants to have the field XRD_MB_QPSA_TERM populated regardless of marital status
- However, with recent requirements to value all deferred participants as Unknown marital status, the second reason is not as important.
 - A plan that has the same QPSA and QJSA may not find many benefits from the addition of the QPSA row

QPSA Row

- Programming the QPSA row
 - Output fields in the QPSA row should be Input fields in the XRD row
 - In most cases, the QPSA row is a copy of the XRD row with the 'O' and 'B' values in the XRD I/O/B row changed to 'N' in the QPSA I/O/B row. The two exceptions are XRD_MB_QPSA_TERM and LS_QPSA which would generally be an 'O' in the QPSA I/O/B row and an 'I' in the XRD I/O/B row

QPSA Row

- Programming the QPSA row
 - In cases where the QPSA does not equal the QJSA, additional replacements can be made to calculate the benefit in the QPSA row:

<i>XRD I/O/B</i>	B	B	B
XRD Field name	FORM_CODE	SPC_VAL	MTHS
<i>QPSA I/O/B</i>	I	I	I
QPSA Field name	FORM_CODE_PTP_QPSA	SPC_PTP_QPSA	MTHS_PTP_QPSA

- The QPSA monthly benefit will be calculated in the output field XRD_MB_QPSA_TERM on the QPSA row then pulled onto the XRD row. Valuation QPSA liabilities should be based on this value

QPSA Row

- Programming the QPSA row
 - Care should be taken when using the QPSA row, particularly for complex QPSA calculations where the QPSA does not equal the QJSA. DRS Listings will need to incorporate fields in the QPSA row as well as the XRD row. These fields will likely need to be user defined fields instead of 'N' in the QPSA I/O/B row.

RASD Row

- RETRO_DOR replaces XRD as an input
- Calculated in the NSF unless there is a post-DOPT death or deferred beneficiary
- Limited pre-defined RASD fields.
Must assign user defined fields for additional fields needed for DRS and statements

XRD Field	RASD Field
XRD	RETRO_DOR
FORM_CODE	FORM_CODE_NSF
SPC_VAL	SPCNSF
MTHS	MTHSNSF
ANB_XRD	N/A
SANB_XRD	N/A
BFCF_XRD	N/A
ME_XRD	N/A
ERF_XRD	N/A
ML_XRD	N/A
LRF_XRD	N/A
XRD_PMB	N/A
CASH_BAL_XRD	N/A
CASH_BAL_FACT_XRD	N/A
MAXLIM_XRD	N/A
XRD_MB_TITLE_IV	RETRO_MB_TITLE_IV
XRD_MB_4022C	RETRO_MB_4022C
XRD_MB_TERM	RETRO_MB_TERM
XRD_SURV_MB_TERM	RETRO_SURV_MB_TERM

NRD/EURD/ERD Row

- Replace XRD fields with equivalent NRD/EURD/ERD field

XRD Field	NRD Field	EURD Field	ERD Field
XRD	NRD	BDEUAS	BDERDLEV
FORM_CODE	FORM_CODE_NSF	FORM_CODE_NSF	FORM_CODE_NSF
SPC_VAL	SPCNSF	SPCNSF	SPCNSF
MTHS	MTHSNSF	MTHSNSF	MTHSNSF
ANB_XRD	NRA	ANB_EURD	ANB_ERD
SANB_XRD	SNRA	SANB_EURD	SANB_ERD
BFCF_XRD	BFCF_NRD	BFCF_EURD	BFCF_ERD
ME_XRD	N/A	N/A	ME_ERD
ERF_XRD	N/A	N/A	ERF_ERD
ML_XRD	N/A	ML_EURD	ML_ERD
LRF_XRD	N/A	LRF_EURD	LRF_ERD
XRD_PMB	N/A	EURD_PMB	ERD_PMB
CASH_BAL_XRD	CASH_BAL_NRD	CASH_BAL_EURD	CASH_BAL_ERD
CASH_BAL_FACT_XRD	CASH_BAL_FACT_NRD	CASH_BAL_FACT_EURD	CASH_BAL_FACT_ERD
MAXLIM_XRD	MAXLIM_NRD	MAXLIM_EURD	MAXLIM_ERD
XRD_MB_TITLE_IV	NRD_MB_TITLE_IV_NSF	EURD_MB_TITLE_IV_NSF	ERD_MB_TITLE_IV_NSF
XRD_MB_4022C	NRD_MB_4022C_NSF	EURD_MB_4022C_NSF	ERD_MB_4022C_NSF
XRD_MB_TERM	TERM_MB_NRD_NSF	TERM_MB_EURD_NSF	TERM_MB_ERD_NSF
XRD_SURV_MB_TERM	TERM_SURV_MB_NRD	TERM_SURV_MB_EURD	TERM_SURV_MB_ERD

NRD/EURD/ERD Row (continued)

- Calculated fields that are not needed can be changed to an “N” on the I/O/B Row
- Calculated in the NSF unless there is a post-DOPT death or deferred beneficiary

DORNSF/DOR Row

- Used to calculate benefits for all post-DOPT retirees at ARD
- DORNSF will be used to calculate benefits at ARD in the NSF including single PACS optional forms
- DOR will be used to calculate benefits at ARD in the final form
- Use plan factors for plan optional forms for participants put into pay by the PPA on and before DOTR (or completed an application prior to DOTR)

XRD Field	DOR Field
XRD	DOR
FORM_CODE	FORM_CODE_ARD
SPC_VAL	SPC_ARD
MTHS	MTHS_ARD
ANB_XRD	ARA
SANB_XRD	SARA
BFCF_XRD	BFCF_ARD
ME_XRD	ME_ARD
ERF_XRD	ERF_ARD
ML_XRD	ML_ARD
LRF_XRD	LRF_ARD
XRD_PMB	ARD_PMB
CASH_BAL_XRD	N/A
CASH_BAL_FACT_XRD	N/A
MAXLIM_XRD	MAXLIM_ARD
XRD_MB_TITLE_IV	ARD_MB_TITLE_IV
XRD_MB_4022C	ARD_MB_4022C
XRD_MB_TERM	TERM_MB_ARD
XRD_SURV_MB_TERM	TERM_SURV_MB_ARD

VAL_SDOB at Different Dates

- XRD

- For valuation

- ID = 1, +/- 4 years age assumption
 - ID = 2, SDOB

- For BSRS

- BDOB or SDOB, if available, else blank

- QPSA

- SDOB, if available, else +/- 4 years age assumption

VAL_SDOB at Different Dates (continued)

- RASD/NRD/EURD/ERD
 - SDOB if DOD >0, else blank
- DORNSF
 - Blank
- DOR
 - BDOB or SDOB, if available, else blank

Questions?

Advanced Plan Benefit Programming

Auxiliary Disability

- DOR, FORM_CODE_ARD, SPC_ARD, MTHS_ARD, and ANNUITY_TYPE should be reserved for the final election
- AUX_DIS_COMMENCE_DATE, AUX_DIS_BEN should be populated with the commencement date and amount of the plan's disability benefit
- AUX_DIS_CHANGE_DATE is the expected commencement date of the non-disability benefit, usually the same as NRD
- RETTYP should be set to "DS" (disabled without Social Security) or "DR" (disabled with social security)
- See [Requirements for Participants Receiving Auxiliary Disability Benefits at DOPT](#)

Auxiliary Disability (continued)

- QPSA
 - If a married participant receiving an auxiliary disability benefit dies before commencing the non-disability benefit, the dates in BDERDLEV and BDEUAS should be based on the earliest commencement date of the QPSA instead of on NRD.
 - However, LS_TERM and LS_QPSA must be calculated based on the benefits that commence at the AUX_DIS_CHANGE_DATE.

Auxiliary Disability (continued)

- Present Value Factors for Liabilities
 - There will need to be two annuity PVFs to calculate liabilities
 1. Immediate PVF from DOPT to AUX_DIS_CHANGE_DATE with a temp period
 2. Deferred PVF from DOPT to AUX_DIS_CHANGE_DATE

QDROs

- AP_VAL_IND
 - This field indicates whether the alternate payee (AP) is included in the plan population.
 - APs with AP_VAL_IND set to 'N' should not be included in plan population counts when determining liabilities, load or average weighted ages.
 - APs with AP_VAL_IND set to 'N' should still be included in LEO and the plan's data environment for benefit statement purposes.

QDROs (continued)

QDRO Type	QDRO Effective Date	Participant Status at DOPT	Alternate Payee Status at DOPT	AP_VAL_IND	Alternate Payee's RETSTAT
Separate Interest	Pre DOPT	In Pay / Deceased	In Pay	Y	1
Shared Payment	Pre DOPT	In Pay	In Pay	N	1
Separate Interest	Pre DOPT	In Pay / Deceased	Not In Pay	Y	2
Shared Payment	Pre DOPT	In Pay	Not In Pay	N	2
Separate Interest	Pre DOPT	Not In Pay	In Pay	Y	1
Separate Interest	Pre DOPT	Not In Pay	Not In Pay	Y	Participant's RETSTAT
Separate Interest	Post DOPT	Not In Pay	Not In Pay	N	Participant's RETSTAT
Shared Payment	Any	Not In Pay	Not In Pay	N	Participant's RETSTAT

QDROs

- Splitting the Benefit
 - PA_AMB, AMB, and VB should contain the original unsplit accrued benefit
 - Calculate the split benefits in UD fields:
 - Depending on the situation, the AP's portion can be AP_FIXED_AMT , $AMB_SPLIT_DATE * AP_PERCENT$, $VB * \text{marital portion}$, etc.
 - The participant's portion will be $VB - \text{AP's portion}$.
 - In XRD_PMB, use the split benefit and apply ERF/LRF/BFCF based on the participant age. The AP will also need an age adjustment factor applied
 - If $AP_VAL_IND = "N"$, XRD_PMB should not be split. XRD_MB_TERM and LS_TERM is where the split should occur.

QDROs – Valuation

- For ID=4, be sure that PVF_LEV_ANN and PVF_LEV_LS have the Sex, PAge, and PXRA inputs based on the AP's gender and age
- For RETSTAT <> "1" and ID=4, FORM_CODE, SPC_VAL, and MTHS should be based on the normal single form. For ID=1, they should be based on the normal married form.
- For AP_VAL_IND=Y, verify that all present values are based on the split benefit

QDROs (continued)

- QDROs Effective After DOPT: XRD Assumptions and Liabilities
 - The AP's XRD is determined as of DOPT, not as of the effective date of the QDRO, and is generally the same as the participant's XRD (unless the AP is prohibited from commencing until a date after the participant's XRD.)
 - Example:
 - DOPT: 06/30/2014
 - Participant's XRD: 08/01/2015
 - QDRO Effective Date: 01/01/2016
 - AP's XRD: 08/01/2015
 - We exclude the AP from the valuation population (AP_VAL_IND = N), and the liability is based on the gross XRD benefit before the QDRO adjustment and included in the participant record only.

QDROs (continued)

- Lump Sum Calculations

- Separate Interest

- The preferred method to calculate lump sums is to compute the lump sum value for both the participant and the AP using the XRD benefit payable to each per the terms of the QDRO, regardless of whether the QDRO is executed before or after DOPT.
 - Alternatively, if AP_VAL_IND = “N”, LS_TERM for the participant can be based on the participant’s total benefit and the AP’s LS_TERM will be \$0

- Shared Payment

- The entire lump sum gets valued under the participant’s record

QDROs (continued)

- Marital Portion Calculation
 - QDROs can specify a “community share” or “marital fraction” of the participant’s benefit the AP is entitled, generally by a certain date; e.g. “The alternate payee is entitled to 50% of the participant’s benefit earned as of June 12, 2013.”
 - The QDRO may specify the exact “marital fraction” or the amount of the “community share”. If not, the fraction must be calculated based on the available information.
 - It is important that all available QDRO data is accurately stored in the data environment.

QDROs (continued)

- Marital Portion Calculation

Field	Value
QDRO_PROV_DATE	The date the QDRO was signed or filed.
QDRO_SPLIT_DATE	The end date to determine the marital portion as stated in the QDRO
AP_PERCENT	The percentage payable to the AP as stated in the QDRO.
AP_FIXED_AMT	A fixed dollar amount payable to the AP as stated in the QDRO.
AMB_SPLT_DATE	The accrued monthly benefit calculated as of QDRO_SPLIT_DATE.
PBGC_AP_PERCENT	PBGC's calculated percentage payable to the AP.



QDROs (continued)

- Marital Portion Calculation Samples

QDROs (continued)

- Early Retirement Subsidy
 - QDROs may specify that a portion of an early retirement subsidy that is payable to the participant may be payable to the AP.
 - If the QDRO is silent on the issue, then we do not pay the AP any portion of the early retirement subsidy.
 - If the plan does not pay an early retirement subsidy (in the form of early unreduced retirement benefits), then the QDRO specifications are moot.



QDROs (continued)

- Age Adjustment Factor
 - This only applies to separate interest type QDROs.
 - We apply a factor to the AP's monthly benefit account for the difference in ages of the participant and AP. The factor is based on the life expectancy of the AP.
 - If there is plan practice available for pre-DOPT QDROs, use the same assumptions to calculate post-DOPT age adjustment factors, and round to the same decimal places.
 - If no plan practice exists, calculate the factor as a ratio of NPVFs using the plan's assumptions for actuarial equivalence. If actuarial equivalence is not defined, use PBGC PACS mortality and interest (mortality table AA20 and 6% interest). Round factors to four decimal places.



Pre-Termination Plan Liabilities

- Use PACSIIF function to calculate accumulation factors including interest from DOR to DOPT
 - PACSIIF(StartDate, ThruDate, EstIndicator, LastDate)
 - StartDate should be DOR
 - ThruDate should be first of the month preceding DOPT since that's the last pre-termination payment
 - EstIndicator and Last Date are optional parameters and not needed for this calculation
 - Note: Although ThruDate is first of the month preceding DOPT, interest will be accumulated through one month after ThruDate

Plan Offsets

- See March 2020 Offsets Training

EECs

Employee Contributions (EEC)

See example below to compute the offset for the withdrawal of EEC's:

Compute a PVF as of DOPT for the deferred NRD benefit.

- DOPT - 06/01/2015 & ANB - 55
- NRD - 07/01/2025 & NRA - 65
- Normal Single Form - SLA
- Stability Period is Plan Year, Look Back is 2 Months, and Plan Anniversary is 01/01

EECs

Employee Contributions (EEC)

Compute a PVF as of DOPT for the deferred NRD benefit.

- NPVF2("MP","SLA","L01012",06/15/2015,"M",55,65,99,0,0,0,07/01/2025,07/01/2025,"N","", "")
- Divide the EEC balance at DOPT by the deferred factor computed above to obtain the NRD benefit
- Adjust the NRD benefit at DOPT to XRD by multiplying by the applicable plan ERF and BFCF

EECs

Employee Contributions (EEC)

- To compute the EEC balance at XRD, we determine the segment in which NRD falls
- In the example above this would be the second segment
- Use that rate to accumulate the EEC's to XRD. The EEC balance at XRD is only used for the post-retirement death benefit due to the refund provision provided by the plan.



Lump Sum Reconciliation

- PBGC
 - Note: LS_EST_AMT is often initially populated with the amount paid with interest. You will have to identify the amount without interest for reconciliation.
- Plan
 - Discount plan lump sums to DOPT and compare to LS_TERM

Residual Lump Sums

- When the PBGC estimated lump sum is an underpayment, add the following fields:
 - LS_TERM_RESIDUAL = the amount of LS_TERM left over after subtracting LS_EST_AMT
 - A UD field containing the ratio between the estimated lump sum and actual LS_TERM. Round to 6 digits.
 - XRD_MB_TERM_RESIDUAL = $XRD_MB_TERM * (1 - \text{lump sum ratio})$
 - Also add UD fields on statement lines

Residual Lump Sums (Continued)

- add the following fields (continued):
 - $\text{XRD_SURV_MB_TERM_RESIDUAL} = \text{XRD_MB_TERM_RESIDUAL} * \text{survivorship percent}$
 - UD fields on the statement lines are only necessary for ID=1 QPSA BD statements.
 - When the QPSA form is different from the QJSA, add a UD field for the spouse's QPSA benefit.
 - UD fields on the statement lines are only necessary for ID=1 QPSA BD statements.
 - Consider adding $\text{XRD_MB_QPSA_TERM_RESIDUAL}$, but generally not necessary. When a QPSA line is used, XRD_MB_QPSA_TERM is only calculated on the QPSA line and has no use in BSRS.

Residual Lump Sums (Continued)

- It is good practice to populate the Residual fields with the full benefit when no estimated lump sum was paid.
- Update the formulas in the PACS section to reference XRD_MB_TERM_RESIDUAL instead of XRD_MB_TERM. For ID=1 QPSA benefits, update the PACS section to reference XRD_SURV_MB_TERM_RESIDUAL instead of XRD_SURV_MB_TERM.
- BR_IND should be set to Y when the estimated lump sum is an underpayment. BR_IND should be N for overpayments and matches.
- Note: residual fields will also be used for net lump sums for plans with EEC



Lump Sum Reconciliation (continued)

- Residual Lump Sum Example Calculation
 - The participant's plan monthly benefit at NRD in the NSF is \$50. The participant received an estimated PBGC lump sum of \$3,000. The final valuation lump sum is actually \$3,400. What is the participant's residual monthly benefit at NRD in the NSF?
 1. Residual Ratio: $\$3,000 \text{ (LS_EST_AMT)} / \$3,400 \text{ (LS_TERM)} = 88.2353\%$
 2. Offset for LS_EST_AMT: $\$50.00 \times 88.2353\% = \44.12
 3. Residual MB: $\$50.00 - \$44.12 = \$5.88$



Questions?

Advanced Guaranteed Benefit Programming

Application of Multiple PBGC Limits

- Priority Category Overview:
 - PC1 – Voluntary Employee Contributions
 - PC2 – Mandatory Employee Contributions
 - PC3 – Benefits in pay or could have been in pay 3 years prior to DOPT/BPD, based on provisions providing lowest benefit within 5 years prior to DOPT/BPD
 - PC4 – Guaranteed Benefits (Aggregate Maximum and Majority Owner)
 - PC5 – Non-forfeitable benefits limited in PC4 by BPD, AAN, MIL, or PI
 - PC6 – All benefits not in PC1-PC5

Application of Multiple PBGC Limits

- PBGC Limitations to be applied in order: BPD, AAN, MIL, Phase-In, Majority Owner
- After PC3, benefits affected by these limitations are in PC5
 - PC5 is divided into subcategories for the 5 year period prior to DOPT (not BPD)
 - Should only need subcategories with Phase-In
 - Five year old plan first (PC5A)
 - Amendment by amendment for PC5B, PC5C, etc.
 - Later of the effective date and the adoption date for each increase

Application of Multiple PBGC Limits Example

Consider a Plan with limitations from BPD, AAN, and either MGB or PI

Applying the limits in the proper order, the guaranteed benefit must be determined before allocating the UNGB

1. Calculate Plan Benefit with supplement as of XRD (applying all necessary factors)
2. Calculate Benefit as of BPD with supplement as of XRD
3. Accrued at Normal is equal to the Accrued Monthly Benefit at NRD
 - Determine the guaranteed supplement: Greater of the supplement and the difference between the BPD benefit as of XRD (BPD_XRD_PMB) and the AMB



Application of Multiple PBGC Limits: Level Life Factors

- We now have the monthly benefit at BPD and the Supplement limited to the AAN
- MGB and/or Phase-In cannot be applied to a non-level benefit, and so a level benefit must be calculated using Level Life Factors
- PBGCLLF:
 - DOPT
 - DOB
 - SDOB
 - BDOB
 - NONSPIND
 - ID
 - StartDate
 - EndDate
 - BPD

Application of Multiple PBGC Limits: Level Life Factors

- As of August 2017, the function PBGCLLF replaced LLF (August 2017 ACT User Group Meeting)
- Converting a non-level benefit to one that's level must be done with PV's: the LLF function does this in one step.
- Level Life Benefit = Lifetime Monthly Benefit + [Supplement x LLF]
- Common Mistake: If the additional benefit is a deferred “step up” instead of a temporary supplement, take $(1 - \text{LLF})$, with the End Date being when the “step up” occurs.

Application of Multiple PBGC Limits Example

- Apply LLF to the Guaranteeable supplement in order to get a Level Life Benefit
- Apply the Maximum Limit by calculating a Guaranty Ratio
 - Guaranty Ratio = $\text{MIN}[1, \text{MGB} / \text{Level Life Benefit}]$
- Multiply Guaranty Ratio by BPD Monthly Benefit at XRD and Guaranteeable Supplement for the Guaranteed Monthly Benefit and Supplement (without PC3 or more limitations, this is PC4_LEVEL_MB and PC4_SUPP1_MB)



Application of Multiple PBGC Limits Example

- If Phase-In is to be applied, the benefit for each amendment (DOPT-5, DOPT-4, etc.) must be levelized with the supplement and compared to the MGB before finding the guaranteed increase
- Again calculate Guaranty Ratio, set equal to the Guaranteed Level Benefit (after increases and MGB) divided by the Level Benefit at DOPT

Lump Sum Death Benefits and 4022(c)

- The monthly benefits in PC6 for lump sum death benefits are derived from present values
- By using a PVF calculated at different dates, the monthly benefit at ERD may be greater than the monthly benefit at NRD.
- To avoid this, best practice is as follows
 - Calculate monthly benefit in valuation form at XRD
 - Divide monthly benefit at XRD by ERF_XRD , LRF_XRD , and $BFCF_XRD$ to get monthly benefit at NRD in NSF
 - Use monthly benefit at NRD in NSF as an input at other dates and apply the corresponding ERF , LRF , and $BFCF$

Lump Sum Death Benefits and 4022(c) (continued)

N105	N106	PC6_LSDB_PVMB	PC6_LEVEL_MB	N094
\$1,500	0	\$1,045	\$5.63	\$3.68
			#DIV/0!	
	$A_{XRA} = 1 - d\ddot{a}_{XRA}$	$1,500A_{XRA}$ PV With Load and QPSA	$1,500A_{XRA}/12^{*} \text{XRD-DOPT} \ddot{a}_{Xrd}$	
O	O	O	O	B
N105	N106	PC6_LSDB_PVMB	PC6_LEVEL_MB	N094
N105	N106	PC6_LSDB_PVMB	PC6_LEVEL_MB	N094
N105	N106	PC6_LSDB_PVMB	PC6_LEVEL_MB	N094
N105	N106	PC6_LSDB_PVMB	PC6_LEVEL_MB	N094
N105	N106	PC6_LSDB_PVMB	PC6_LEVEL_MB	N094
N105	N106	PC6_LSDB_PVMB	PC6_LEVEL_MB	N094
N105	N106	PC6_LSDB_PVMB	PC6_LEVEL_MB	N094
N105	N106	PC6_LSDB_PVMB	PC6_LEVEL_MB	N094
N105	N106	PC6_LSDB_PVMB	PC6_LEVEL_MB	N094
Amount of Post-DOR Lump Sum Death Benefit	NPVF for Lump Sum Death Benefit	Pesent Value of Lump Sum Deth Benefit	PC6 Monthly Benefit	PC6 UNGB at NRD
O	O	O	O	B
N105	N106	PC6_LSDB_PVMB	PC6_LEVEL_MB	N094

– PC6_LSDB_PVMB = N105 x N106

$$-PC6_LEVEL_MB = \frac{PC6_LSDB_PVMB}{(N106 \times 12)}$$
$$-N094 = \frac{PC6_LEVEL_MB}{(ERF_XRD \times LRF_XRD \times BFCF_XRD)}$$

BUCK

Lump Sum Death Benefits and 4022(c) (continued)

N095	PC6_4022C_AR	XRD_MB_4022C
\$5.62	8.45%	\$0.47
O	O	O
N095	PC6_4022C_AR	XRD_MB_4022C
O	I	O
N096	PC6_4022C_AR	NRD_MB_4022C_NSF
O	I	O
N097	PC6_4022C_AR	EURD_MB_4022C_NSF
O	I	O
N098	PC6_4022C_AR	ERD_MB_4022C_NSF
O	I	O
N099	PC6_4022C_AR	RETRO_MB_4022C
O	I	O
N103	PC6_4022C_AR	ARD_MB_4022C
O	I	O
N103	PC6_4022C_AR	ARD_MB_4022C
Total PC6 MB adjusted for date and form	PC6 4022(C) Allocation Ratio	4022C Monthly Benefit at XRD
O	O	O
N095	PC6_4022C_AR	XRD_MB_4022C

- PC6 Monthly Benefits adjusted for form and date
 - N095, N096, N097, N098, N099, and N103
 - = N094 x ERF x LRF x BFCF
- 4022(c) Monthly Benefits adjusted for form and date
 - PC6 Monthly Benefits adjusted for form and date x PC6_4022C_AR

Questions?

Checking and Debugging your program

Common Mistakes

- Check lines other than XRD to make sure fields and I/O/B/N/P indicators have all been filled in properly
- Make sure you aren't accidentally overwriting a field at XRD because the same field is being calculated on a different line
- Check to make sure formulas reference the correct row
- Check to make sure BFCFs are calculating for deferred beneficiaries and post-DOPT deaths with known spouses on lines for statements
- If you have any field on the DOR line that is intended to overwrite an XRD value, make sure to update the DOR row last after an XRD

BUCK update

ERRORCODE Add-in Function

- ERRORCODE returns the long description for an error incurred by one of the ATPBGC Excel AddIn functions.
- Does not give additional detail on every error code.
 - The parameter for the function is:
 - *ErrorNum* is the error code or a cell reference for a cell in which a function is entered that is returning an error code.

Questions?

Statement Fields

Statement Fields

- BS_IND
- BR_IND
- MAN_STMT_IND
- OFA_INDICATOR
- REASON
 - In large case with a lot of reasons for decrease, make a lookup table of reasons and add an indicator to the database
 - Makes it easy to change and match the case memo

Statement Fields Programming

- BS_IND: Program to be “Y” when
 - The participant is not in pay status
 - The participant was put into pay by the Plan Administrator and their benefit is changing
 - The participant was put into pay by PBGC
- BR_IND: Program to be “Y” when the participant is not in pay, or is given a chance to elect a new retirement date or form

Statement Fields Programming (con't)

- MAN_STMT_IND: Program to “Y” if the BSRS is unable to generate their statement (usually for QDROs or very unique circumstances)
- OFA_INDICATOR: Can be used to turn off OFA calculations if the participant cannot elect a different form of annuity
- REASON: Used in the v1 to program reasons for benefit changes to be used in the statements; BSRS input then only needs to be REASON

Questions?

Index

Index of abbreviations used

- AAN – Accrued-at-Normal
- ANB – Participant's age nearest birthday at DOPT
- AMB – Accrued Monthly Benefit
- ARD – Actual Retirement Date
- ARA – Actual Retirement Age
- BFCF – Benefit Form Conversion Factor
- BL – Benefit Level
- BPD – Bankruptcy Petition Date
- C&C – Certain-and-continuous annuity
- CB – Cash Balance
- DOB – Date of Birth
- ~~BUCK~~ DOD – Date of Death

Index of abbreviations used (continued)

- DOPT – Date of Plan Termination
- DOR – Date of Retirement
- DOTE – Date of Termination of Employment
- EEC – Employee Contributions
- ERA – Earliest Retirement Age
- ERF – Early Retirement Factor
- ERISA – Employee Retirement Income Security Act
- IRC – Internal Revenue Code
- IRS – Internal Revenue Service
- J&S – Joint-and-Survivor annuity

Index of abbreviations used (continued)

- LRA – Late Retirement Age
- LRF – Late Retirement Factor
- MCR – Modified Cash Refund annuity
- NMF – Normal Married Form
- NRA – Normal Retirement Age
- NRD – Normal Retirement Date
- NSF – Normal Single Form
- PC3 – Priority Category 3
- PEP – Pension Equity Plan
- PVF – Present Value Factor
- PVPBL – Present Value of Plan Benefit Liabilities

Index of abbreviations used (continued)

- QDRO – Qualified Domestic Relations Order
- QJSA – Qualified joint-and-survivor survivor annuity
- QPSA – Qualified pre-retirement survivor annuity
- RASD – Retroactive Annuity Starting Date
- SANB – Spouse's age nearest birthday at DOPT
- SDOB – Spouse's Date of Birth
- SLA – Straight Life Annuity
- SSA – Social Security Administration
- SXRA – Spouse's age at participant's XRD
- XRA – Expected retirement age
- XRD – Expected retirement date































